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A Conceptual Structural Health Monitoring System based on Vibration and Wave Propagation

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Development of efficient methodologies to determine the presence, location, and severity of hidden damage in critical structural components is an important task in the design and construction of structural health monitoring systems in aging as well as new structures. In this article, a methodology for automatic damage identification and localization is presented. The structure is assumed to be instrumented with an array of actuators and sensors to excite and record its dynamic response, including vibration and wave propagation effects. In the vibrational approach, the data consist of the modal response of the structure produced by the actuators while in the wave propagation approach, they are the broadband signals due to ultrasonic waves propagating in the structures. Both types of signals are affected by the presence of defects. The approximate location and severity of an unknown defect is determined using a damage correlation index calculated from the frequency response function (FRF) of the structure. The damage index is a relative measure whose value depends on the differences in the dynamical properties of the undamaged (baseline) and damaged structures. The method is applied to simple structural components involving aluminum beams and plates with reduced local stiffness and a composite plate with impact damage. The potential application of the technique to practical health monitoring problems is discussed.

Keywords structural health monitoring · hidden defect · damage index · frequency response · ultrasonic wave · aluminum · composite laminate

1 Introduction

Every engineered structure, no matter how well constructed, contains flaws caused by aging, service loads, or manufacturing processes. For example, laminated composite materials, used in

many aerospace and aircraft structures, often contain a variety of hidden internal flaws, which occur during manufacturing and processing of the material or while the structure is subjected to service loads. A major concern in these structures is the growth of undetected hidden

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damage caused by fatigue and low velocity foreign object impact that can reach a critical size during service, resulting in catastrophic failure of the structure. Damage tolerant and fail-safe design of both new and aging aircrafts, aerospace and civil structures require a substantial amount of inspection and defects monitoring at regular intervals. According to some estimates, over 25% of the life cycle cost of an aircraft or aerospace structure, which includes pre-production, production, and post-production costs can be attributed to operation and support, involving inspection and maintenance. The life cycle cost of new aircraft and aerospace structures can be reduced significantly if continuous and autonomous condition-based structural health monitoring systems can be integrated into their design.

Two general approaches to structural health monitoring are currently being pursued by many investigators. One is the global approach in which changes in the vibrational properties of a structure caused by damage are measured and analyzed in an effort to detect and characterize the damage [1–7]. The other is a local approach, whereby changes in the characteristics of ultrasonic waves propagating across existing defects or created by emerging defects are measured and analyzed to detect and characterize them [8–15]. Both approaches have limitations in their practical implementations, particularly for larger structures. The global approach is only effective in dealing with larger defects for the obvious reason that the effects of small flaws on the global vibrational properties are often below the noise level in large structures. The local approach, although highly effective in detecting very small defects, requires a dense network of sensors that is impractical to implement in larger structures. Other limitations of the current defects detection methodology are the presence of ambient noise and more importantly the need for manual analysis of the signals resulting in extremely time-consuming inspection procedure and strong operator-dependent conclusions. Since the effectiveness of structural health monitoring systems is predicated by its ability to detect small defects before they grow to a critical size, it is imperative that more attention be

focused on the issues limiting their practical utility.

The approach described in this article stipulates the use of a sparsely distributed network of low frequency sensors to measure the global response and clusters of high-frequency ultrasonic sensor arrays located in critical areas of a structure. A unified automated analysis procedure is applied to all measured data to determine the approximate location and the general severity of existing or emerging defects with minimal operator intervention. In most cases, the current nondestructive evaluation (NDE) methods can then be used for precise localization and characterization of the flaws. As an example, flaws parallel to the surface of a structural component (e.g., delamination in a composite laminate) can be very precisely imaged by conventional ultrasonic techniques if their approximate location is known [8–10]. The same is true for many other critical defects in accessible areas of most structures.

2 Vibrational Approach

A great deal of work has been done on the analysis of the global dynamic behavior of structures for damage detection purposes [1,2]. Studies have been conducted to relate the changes in the modal parameters to detect and characterize the damage [4–6] in idealized structures. As an example, the appearance of a notch or surface crack produces a change in the physical properties (e.g., stiffness) of a structure and consequently in the modal parameters. As the frequency response function (FRF) depends on such properties, it is in principle possible to detect and localize the defect by evaluating the changes in the FRF. The presence and extent of damage can be determined by measuring or calculating the FRF of the dynamic response of the structure before and after the damage appears. For each damage size and location, it is possible to define a dimensionless quantity using the undamaged and damaged FRFs. The quantity is referred here as the damage index, D , and is representative of the damage level and its location.

The basic concept behind this damage identification method is illustrated by considering the vibration of an aluminum beam of length 350 mm and cross section $10 \times 3 \text{ mm}^2$ with both ends clamped as shown in Figure 1. The response of the beam to applied loads was calculated using the finite element method (NASTRAN). The presence of damage was simulated by progressively reducing the area moment of inertia of 15% in steps of 5% in one element of the beam, which constitutes 2% of its entire volume. The simulated damage is an approximate representation of thickness loss due to corrosion or service-related structural degradation. The beam was excited by a unit force with a flat spectrum so that the calculated response represents the FRF. The response was calculated at a number of control points (CP) along the beam axis. For each damage level and location, a frequency response analysis was conducted and the displacements and

velocities at the control points were recovered as functions of frequency, as shown in Figure 2. It can be seen that the flaws have very small effects on the modal frequencies of the beam and the variation of the frequencies at the peaks of the FRF (i.e., at the modal frequencies) is smaller than the variation at frequencies far from the modal ones. Thus it would be difficult, if not impossible, to use the modal properties directly to identify damage in the beam. In addition, more information about the structural health can be obtained if the analysis is carried out over the entire frequency range of interest.

To improve the effectiveness of the modal approach, the FRFs of the structure are used to define a damage index, $D_{i,j,DL}$, as follows:

$$D_{i,j,DL} = \left| 1 - \frac{\{R_i\}_{DL}^T \times \{R_j\}_{DL}}{\{R_i\}_{DL=0}^T \times \{R_j\}_{DL=0}} \right| \quad (1)$$

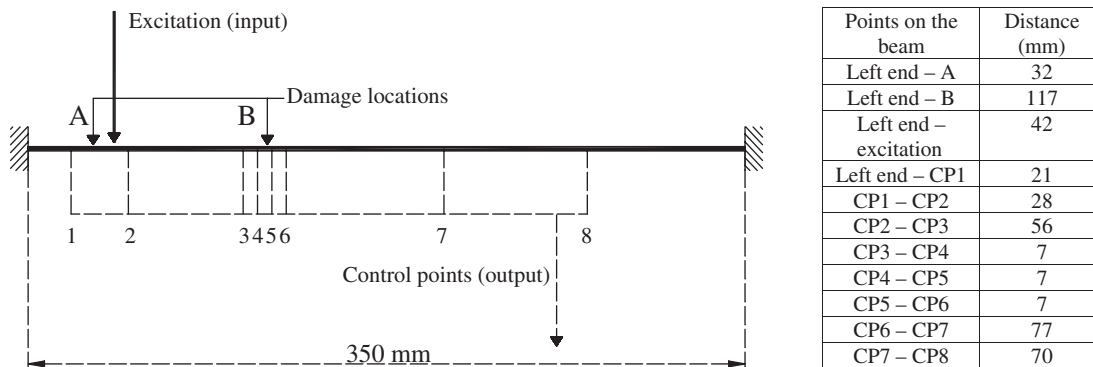


Figure 1 The excitation point, control points, and damage locations (A and B) for the fixed ended aluminum beam. The extension of the damage is 7 mm for both A and B.

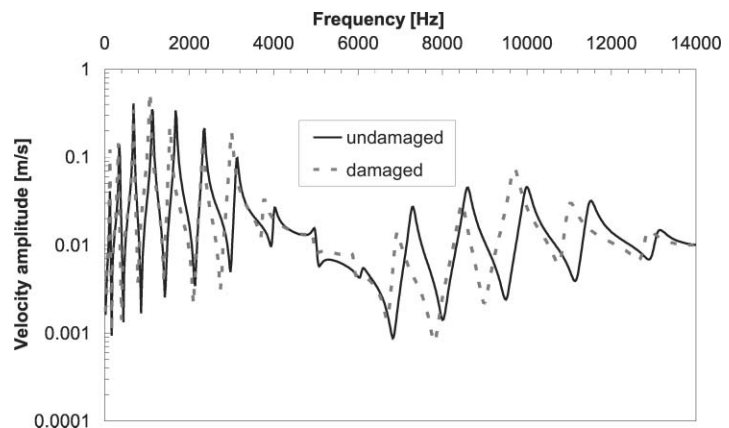


Figure 2 Calculated velocity vs frequency at control point #6 on the beam.

where DL is the damage level (DL=0 for no damage) and $\{R_{ij}\}_{DL}$ is the structural response vector at the control point i for a damage level DL. The elements of $\mathbf{R}$ are structural response parameters calculated in the frequency range of interest. The index can, in principle, be defined for a generic structural parameter including displacement, velocity, acceleration, strain, or voltage measured by embedded or attached sensors. The damage index vanishes for an undamaged structure, i.e., DL=0 and the occurrence of damage results in an increase in the value of the index. The index is denoted by $D1$ for $i=j$, and by $D2$ for $i \neq j$. It should be noted that the damage index is similar to the indices FRIC and FRAC introduced [3]. Clearly, the value of the index increases as the difference in the dynamic response of the damaged and

undamaged structure increases. It follows that the value of the index would be higher in the vicinity of damage.

The changes in the damage indices $D1$ and $D2$ for three simulated damage levels at points A and B of the model in Figure 1 is shown in Figure 3. The velocity, V , was used as a structural response parameter. This choice of the structural response parameter appears to give higher sensitivity of the index to the types of damage studied here, although no attempts were made to carry out a systematic study of the other possible choices. In many applications, the choice will generally be limited by the nature of the measured response, as will be seen later in our wave propagation experiments. It is evident that the indices increase with the level of damage, and more importantly, the increase is more

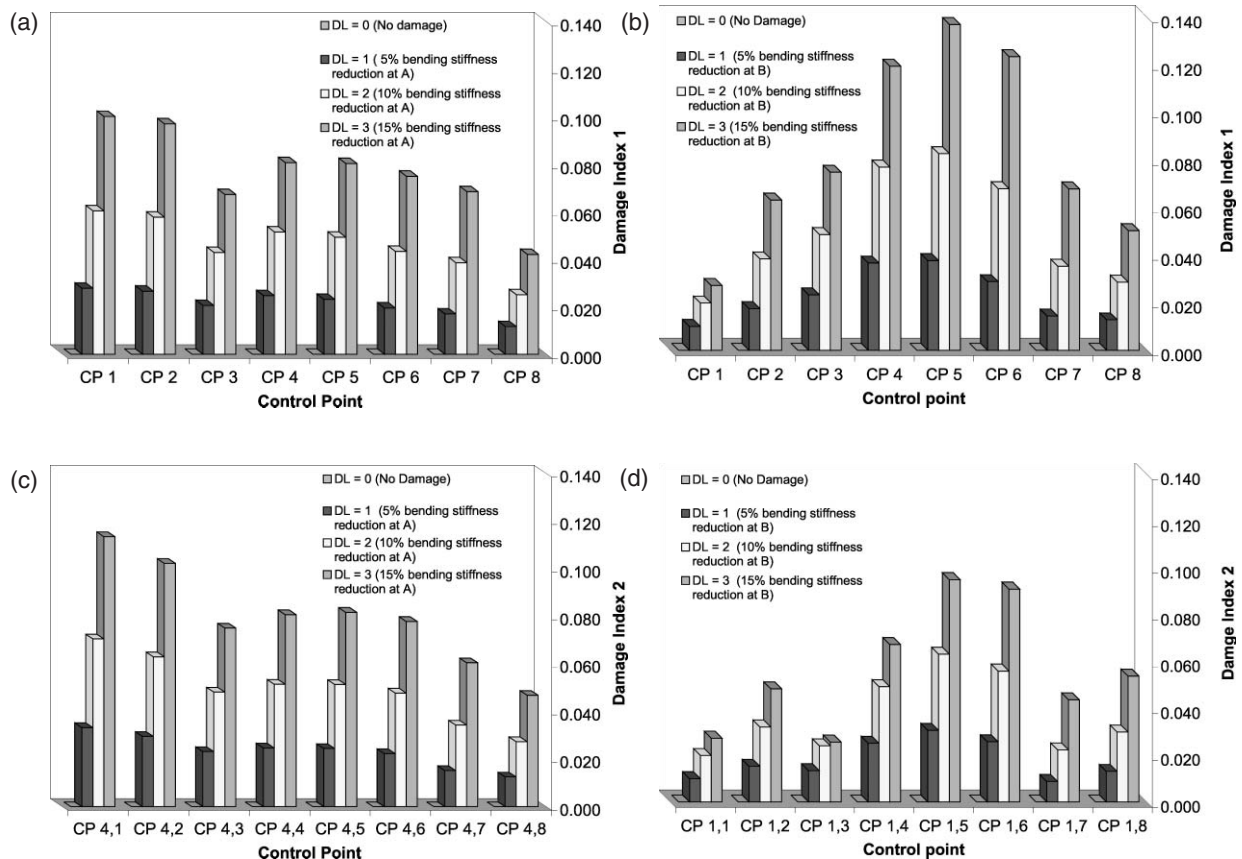


Figure 3 (a) Damage index $D1$ for three damage levels at point A. (b) Damage index $D1$ for three damage levels at point B. (c) Damage index $D2$ for three damage levels at point A. $D2$ was calculated for $i=4$ and $j=1-8$. (d) Damage index $D2$ for three damage levels at point B. $D2$ was calculated for $i=1$ and $j=1-8$.

pronounced at control points closer to the damage location.

The damage index technique is applied next to a simply supported aluminum plate, $600 \times 400 \text{ mm}^2$ in lateral dimensions and 1.5 mm in thickness. The dynamic behavior of the plate was simulated with a finite element model. The damage was introduced into the model by reducing the bending stiffness at two locations. The bending stiffness was progressively reduced up to 10% of the initial value over an area of $20 \times 20 \text{ mm}^2$ at the points A and B shown in Figure 4. Each damage location was analyzed separately as far as the damage index evaluation is concerned.

The velocity V was calculated at nine control points in the frequency range 0–2500 Hz for a unit, flat spectrum excitation and used for the evaluation of the damage index $D1$. The results of the simulations are shown in Figure 5(a)–(f). For both damage locations, the damage index increases with the damage level. Furthermore, the increase is more pronounced at the control points nearer to the damage location.

3 Wave Propagation Approach

The problem of damage identification in an impact-damaged composite plate using a local high frequency transducer array is considered next as an application of the damage index approach.

The test sample was a cross-ply graphite epoxy (AS4/3502) composite laminate with a stacking sequence of $[0/90]_{8s}$ and a thickness of 3.8 mm.

The general experimental setup used for the ultrasonic experiment is sketched in Figure 6(a). Several broadband transducers (B1025, Digital Wave) with 5 MHz center frequency and 6 mm in diameter are placed on the plate surface. Transmission of ultrasound from the transducer to the plate is aided by the application of ultrasonic gel couplant (Sonotech). Due to the variability of the ultrasonic signal strength for different impact events, a set of variable signal attenuators are used to uniformly decrease the signal amplitude so that the signal at the oscilloscope is optimal. The data acquisition module (F4000 Fracture Wave Detector, Digital Wave) is used only as a power source to the preamplifier and the transducers. The AE preamplifiers (PA4020G, Digital Waves) are used as buffers for the recording devices against unexpected spikes in the transducer output. The transducers are connected to the signal attenuators, and are then spliced off from the preamplifiers to be connected directly to a 100 MHz four-channel digital oscilloscope (54624A, Agilent Technologies), where the ultrasonic signal is digitized and recorded. The digital ultrasonic signal can then be downloaded directly to a personal computer for post processing via the data acquisition software IntuiLink (Agilent Technologies). The response of the attenuators and the face-to-face calibration of the

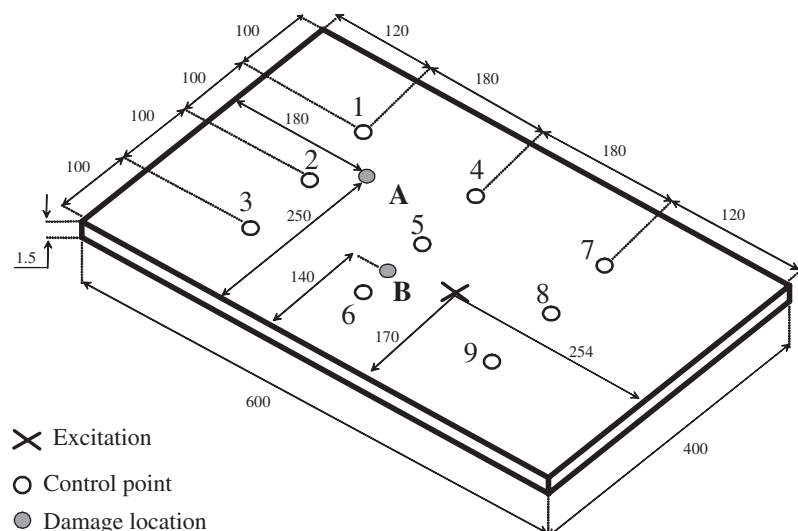


Figure 4 Control points, damage, and excitation locations for the aluminum plate.

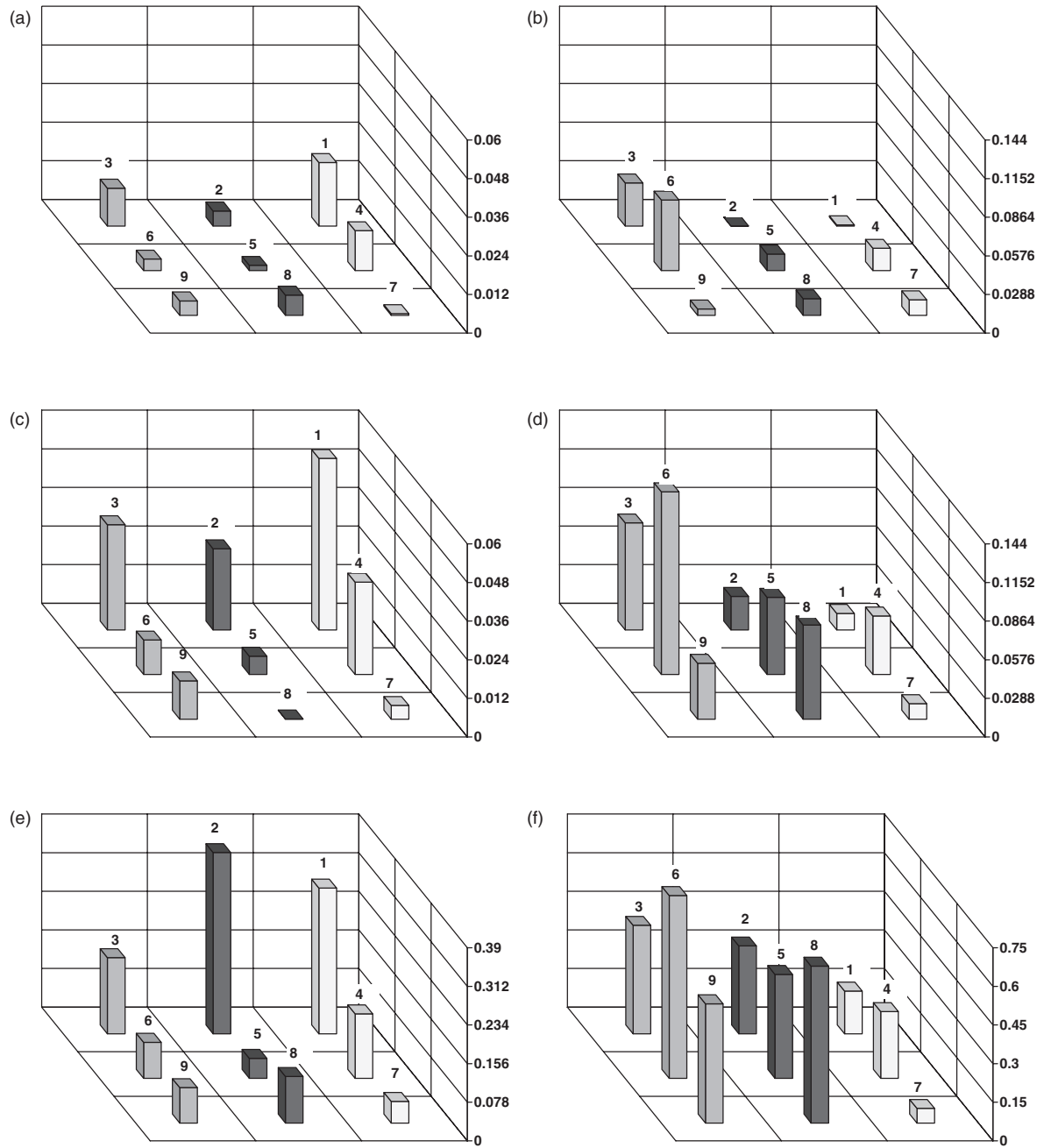


Figure 5 (a) Damage index $D1$ at the control points (30% bending stiffness reduction at A). (b) Damage index $D1$ at the control points (30% bending stiffness reduction at B). (c) Damage index $D1$ at the control points (50% bending stiffness reduction at A). (d) Damage index $D1$ at the control points (50% bending stiffness reduction at B). (e) Damage index $D1$ at the control points (90% bending stiffness reduction at A). (f) Damage index $D1$ at the control points (90% bending stiffness reduction at B).

B1025 transducers with the preamplifiers connected to them are shown in Figure 6(b) and (c). For the frequencies of interest (0.1–1 MHz) the attenuation of the signals is sufficiently uniform

and the phase change in the signal is negligible, demonstrating that the signal recorded with the new setup will not deviate from the original system.

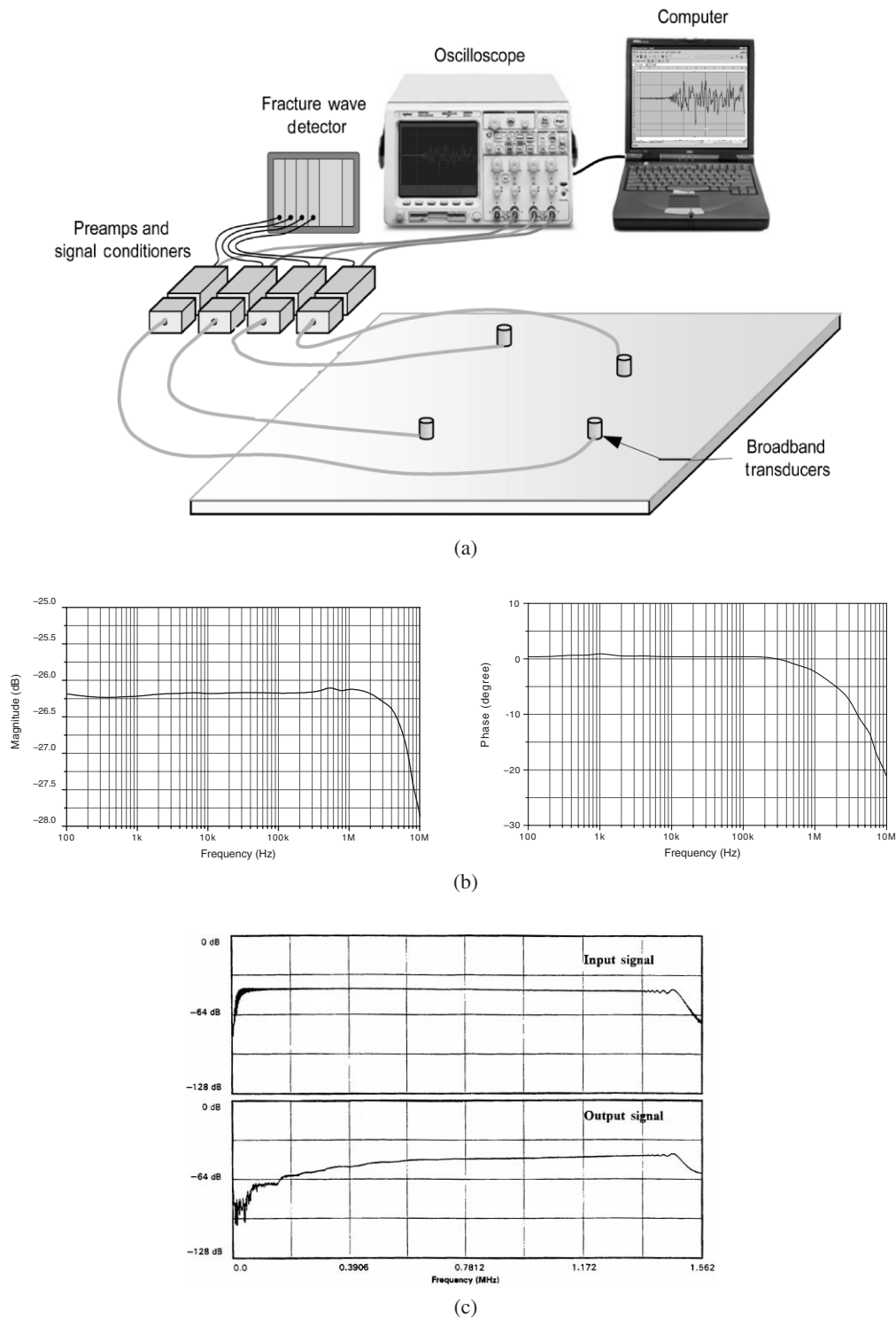


Figure 6 (a) Sketch of the ultrasonic experiment; (b) response of the attenuators; and (c) swept sine wave face-to-face calibration of B1025 sensors.

Wave propagation tests were first performed on the intact plate. The elastic waves were generated by one of the transducers at a point of the plate and the signals were recorded at ten control points. An impact test was then performed using an instrumented drop-weight frame (Instron Dynatup 8250) (Figure 7). The plate was supported by a fixture with a center span of 180 mm. The long edges of the plate were clamped to the fixture while the short ones were free. The impactor (tup) was hemispherical in shape with 12.7 mm in diameter, as shown in the right panel of Figure 7. The load, energy, velocity, and deflection of the impacted point in the plate are shown in Table 1 and Figure 8.

From previous impact tests carried out on the same type of plate at the same impact energy, it has been found that the delamination was

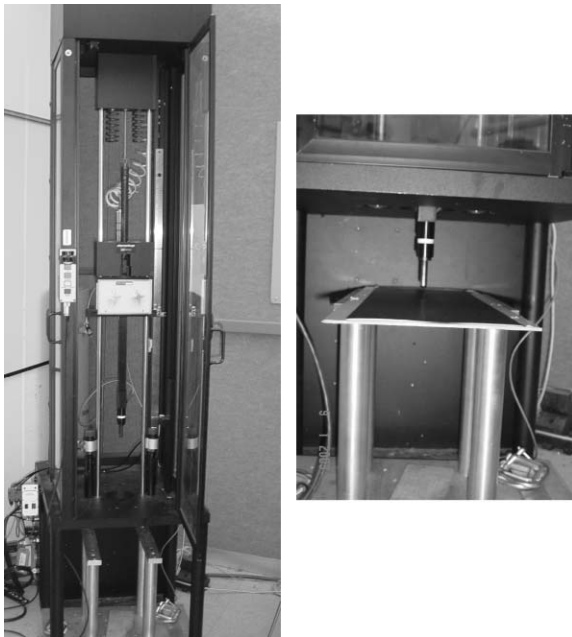


Figure 7 Impact test machine Instron-Dynatup 8250 (left). Tup and fixture details (right).

limited within a circle of 15 mm radius [13]. The ultrasonic wave propagation tests were then repeated as in the intact configuration. The plate and the source/sensor locations are sketched in Figure 9. In both the pre- and post-impact cases, the signals were recorded in the time domain and transformed in the frequency domain (Figure 10), to be used for the computation of the damage index, $D1$, defined earlier.

The elastic waves generated by the source transducer are affected by the impact damage sustained by the laminate. Thus the pre-impact and post-impact signals at all control points differ in their properties. The differences in the signals are caused by reflection, scattering, and diffraction of the waves by the damaged region and careful model-based analysis of the signals can, in principle, be used to determine the nature, location, and severity of the damage. However, the damage caused by the impact is extremely complex and the solution of even the direct problem of wave interaction with the damaged region is not available in the literature. The inverse problem is even more difficult if not impossible to address. Moreover, the modeling approach would require analysis of the differences in the pre- and post-impact signals at each control point, requiring extensive human intervention. A more pragmatic approach is to use the damage index approach to approximately locate the delaminated region and then to use the conventional NDE technique for more refined localization and characterization of the damage (see, e.g., [15]). The results are shown in Figure 11. The delamination modifies the elastic wave path between the source and the receiver at the control points 6 and 7. The changes in the response at these points are well predicted by the damage index evaluated at the ten control points on the composite plate. The index assumes the higher values at control points 6 and 7, as shown in Figure 11.

Table 1 Summary of the main parameters measured during the impact tests.

<i>Impact velocity</i> (m/s)	<i>Maximum load</i> (kN)	<i>Impact energy</i> (J)	<i>Energy to maximum load</i> (J)	<i>Time to maximum load</i> (ms)	<i>Deflection to maximum load</i> (mm)
2.71	6.30	23.03	21.47	3.61	7.52

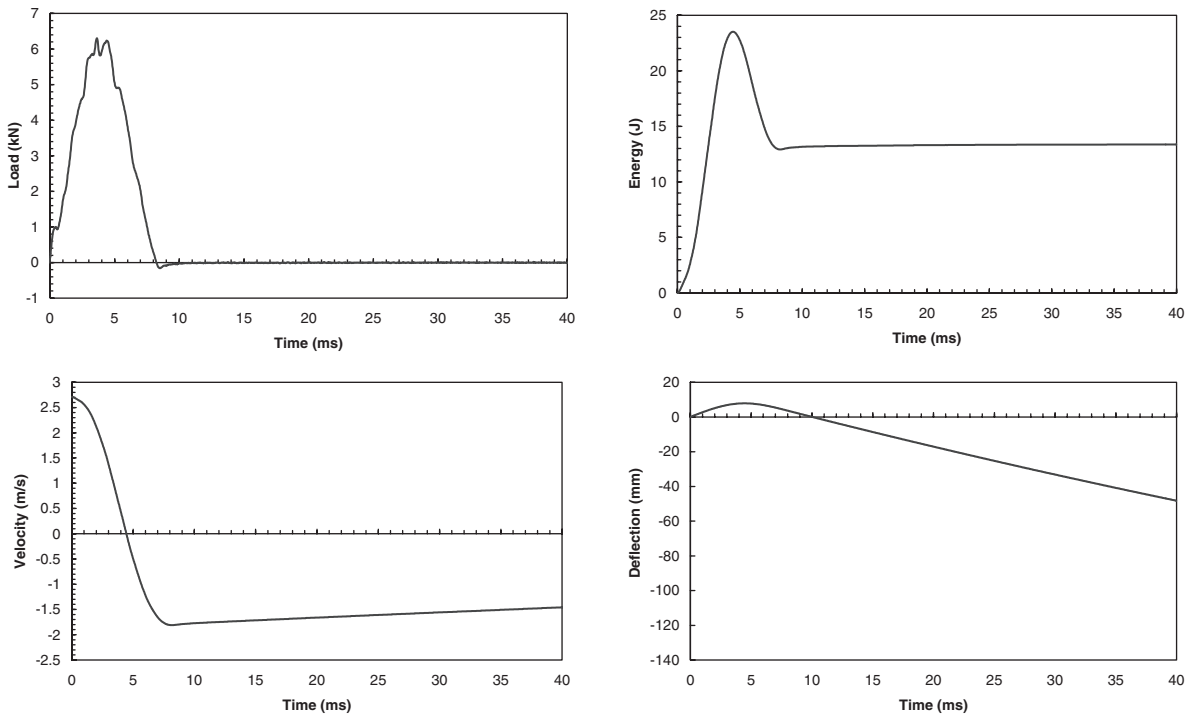


Figure 8 Load, energy, velocity, and deflection time history measured during the composite plate impact test.

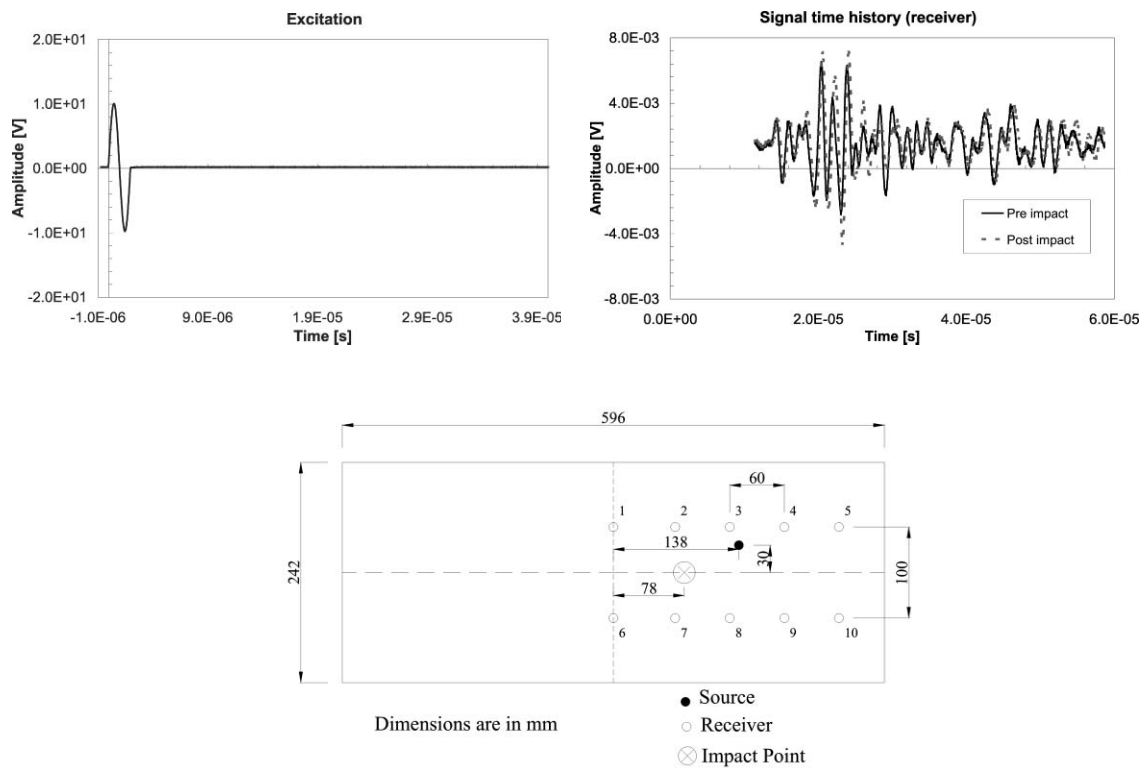


Figure 9 Surface load generated, pre- and post-impact signals recorded in the time domain and source, receiver, and impact point locations on the composite plate.

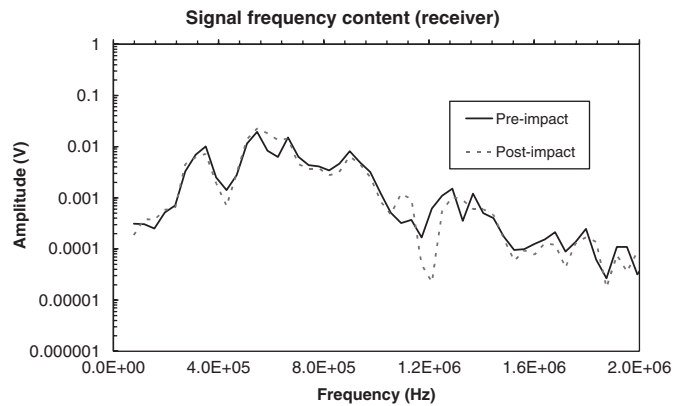


Figure 10 Signals recorded by the receiver at a control point in the frequency domain before and after the impact test.

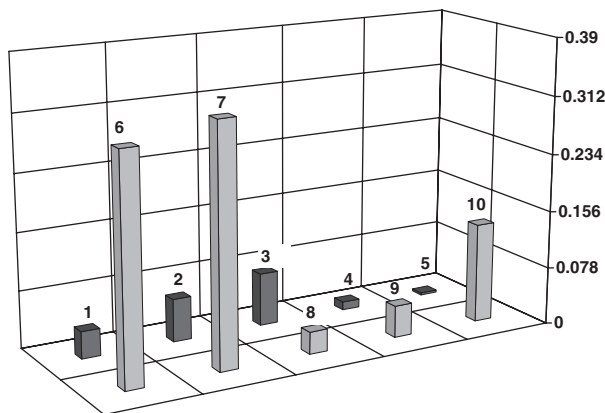


Figure 11 Damage index at the ten control points on the composite plate.

4 Conclusions

The approach presented here can be used for the detection and, in certain conditions, characterization of materials degradation in aircraft, aerospace, and civil structures. For complex structures under realistic service conditions, analysis of the vibrational data is expected to provide information on the existence and the general location of major defects (e.g., widespread damage) within the structure. Analysis of the waveform signals is likely to yield a more detailed information on the location and nature of smaller defects. The unified computer-assisted data analysis procedure should improve the reliability of the defects detection capability and aid in the development of autonomous health monitoring systems for defects-critical structures. It should be

noted that the method proposed here has a number of limitations. These include the inability for accurate localization and detailed characterization of the damage. Some of the limitations may be addressed through further research. As an example, the magnitude of the damage index is related to the reduction in stiffness in the finite beam and plate problems discussed in Section 2. Likewise, the size of the delamination in the large composite plate is related to the magnitude of the damage in Figure 11. It should, in principle, be possible to determine these relationships through a more detailed numerical modeling and experiments with a large number of specimens containing different defects. Moreover, there are few smaller peaks at locations away from the damage, especially, in the modal approach, only near boundaries, which may or may not be present in large structures. Work on the other related issues (e.g., the influence of various types of noise) are currently underway. Nonetheless, the main contribution of the present work is the use of an algorithm that can be applied to large amounts of low frequency vibration data and high frequency wave propagation data to detect and approximately localize defects in structural components without extensive manual intervention. As indicated in the introduction, a more accurate localization and characterization will require detailed local nondestructive testing of the identified areas with possible presence of defects. Even with these limitations, the damage index approach has the potential to significantly reduce the time and cost of inspection of large structures.

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